MIL-V-3144B
6 February 1967
SUPERSEDING
MIL-V-3144A
17 January 1956

MILITARY SPECIFICATION

VIALS, LEVEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers cylindrical and circular level vials used in fire control instruments and similar equipment. (See 6.1).

1.2 Classification.- Level vials shall be of the types specified in figures 1 thru 5.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the supplier shall be as specified in the contract.

3.2 Material.- Materials used shall be capable of withstanding environmental requirements specified herein and show no evidence of being cracked, chipped or peeled.

3.3 Construction.- Vials shall be constructed in accordance with proven commercial techniques and shall conform to the applicable configuration, markings and sensitivity as specified on figures 1 thru 5.

FSC 1290

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3.4 Performance.

3.4.1 Temperature.- When vials are subjected for not less than twenty minutes to the following temperatures in the sequence noted the vials shall meet the performance requirements as specified:

Temperature	Requirement
Boiling water	3.2
Room temperature (60°F to 90°F)	3.2, 3.4.2
-65°F	3.2, 3.4.2, 3.4.2.1
-80°F	3.2, 3.4.3

3.4.2 Coalescence.- When vials are vigorously shaken the bubbles shall coalesce into one large bubble within one second at 60°F to 90°F and within two seconds at -65°F.

3.4.2.1 Circular vial bubble.- In addition to the coalescence requirement of 3.4.2 the bubble in the circular vial shall not be in contact with the base at a temperature higher than -65°F.

3.4.3 Precipitate or solid.- No precipitate or solid shall form within the vial when subjected to a temperature of -80°F.

3.4.4 Room temperature.- Subsequent to 3.4.1 the following requirements shall be met at room temperature (60°F to 90°F).

3.4.4.1 Graduations and markings.- Graduations, level vial type mark and background shall show no signs of being water soluble. The markings shall be free from distortion with clear sharp edges and shall not rub off.

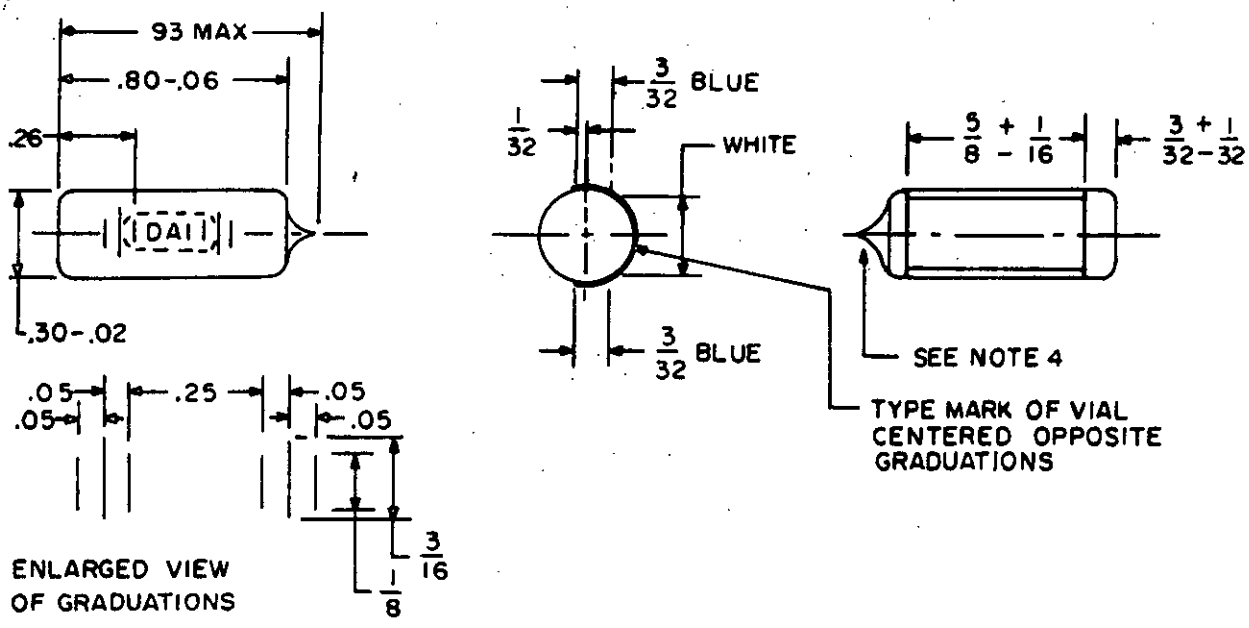
3.4.4.2 Bubble movement and size.- The bubble shall have a uniform movement over the graduated portion of the vial and shall be of the size specified on the applicable figure.

3.4.4.3 Bubble displacement.- When this requirement is applicable (see 6.2) the cylindrical vial shall be rotated about its longitudinal axes through a 90 degree arc, 45 degrees each side of vertical. The amount of out-of-parallel condition between the axis of the inner cylinder and the axis of the outer cylinder shall require the vials to be tilted not more than 2 minutes of angle to return the bubble to its center position.

3.4.4.4 Sensitivity.- The sensitivity shall fall within the range for the type specified on the applicable figure. Vials of sensitivity other than indicated on the figures will be covered by separate drawings.

3.5 Workmanship.- Workmanship shall be in accordance with the best commercial practice for this type of item.

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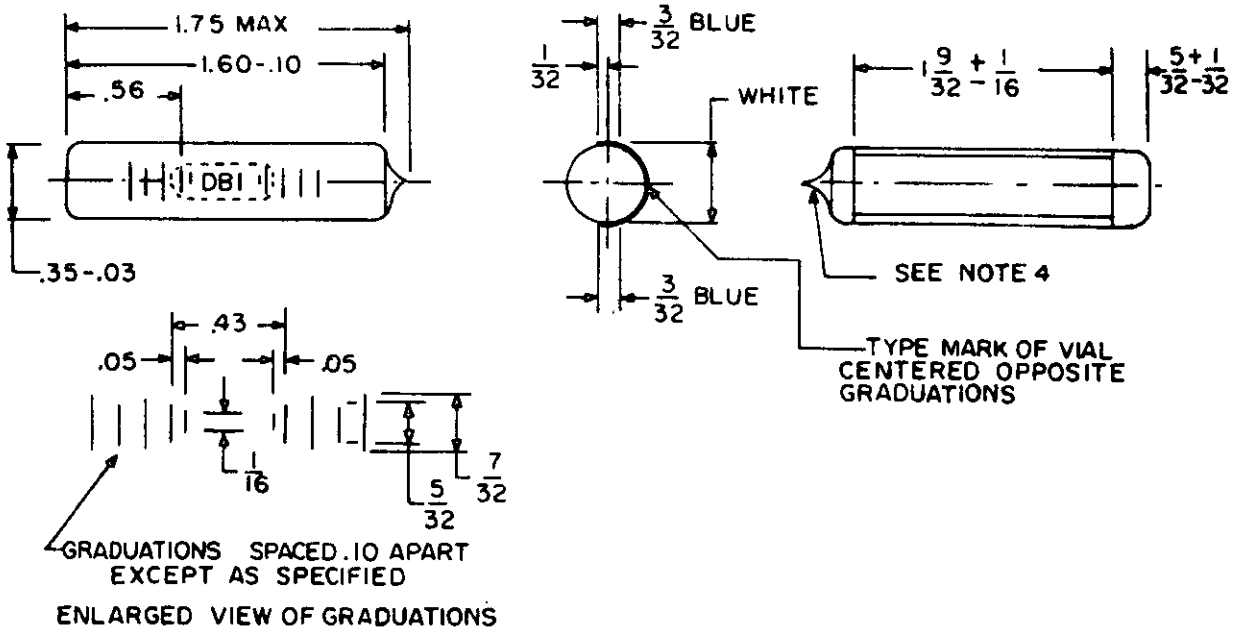
SENSITIVITY. SENSITIVITY OF LEVEL VIALS (BY TYPE) DEFINED IN MINUTES AND SECONDS ("'") OF ARC PER ONE TENTH INCH BUBBLE TRAVEL SHALL BE AS FOLLOWS:

TYPE	SENSITIVITY	TOLERANCE MAX	TOLERANCE MIN
DA1	1'00"	0'40"	1'20"
DA2	2'30"	2'00"	3'00"
DA3	5'00"	4'30"	5'30"
DA4	25'00"	20'00"	30'00"

NOTES:
1. TYPE MARK TO BE 1/16 HIGH, COLOR RED
2. GRADUATIONS TO BE .015 ± .005 THICK TO THE LENGTHS SPECIFIED, COLOR RED
3. BUBBLE LENGTH .350 ± .015
4. CONSTRICTION OF TUBE OPTIONAL. EXTENDED TIP PERMISSIBLE TO .93 MAX OR FLUSH TO .80 DIMENSION
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCE: DECIMAL ± .010, FRACTIONAL ± 1/64

FIGURE 1. VIAL, LEVEL, CYLINDRICAL, TYPE DA

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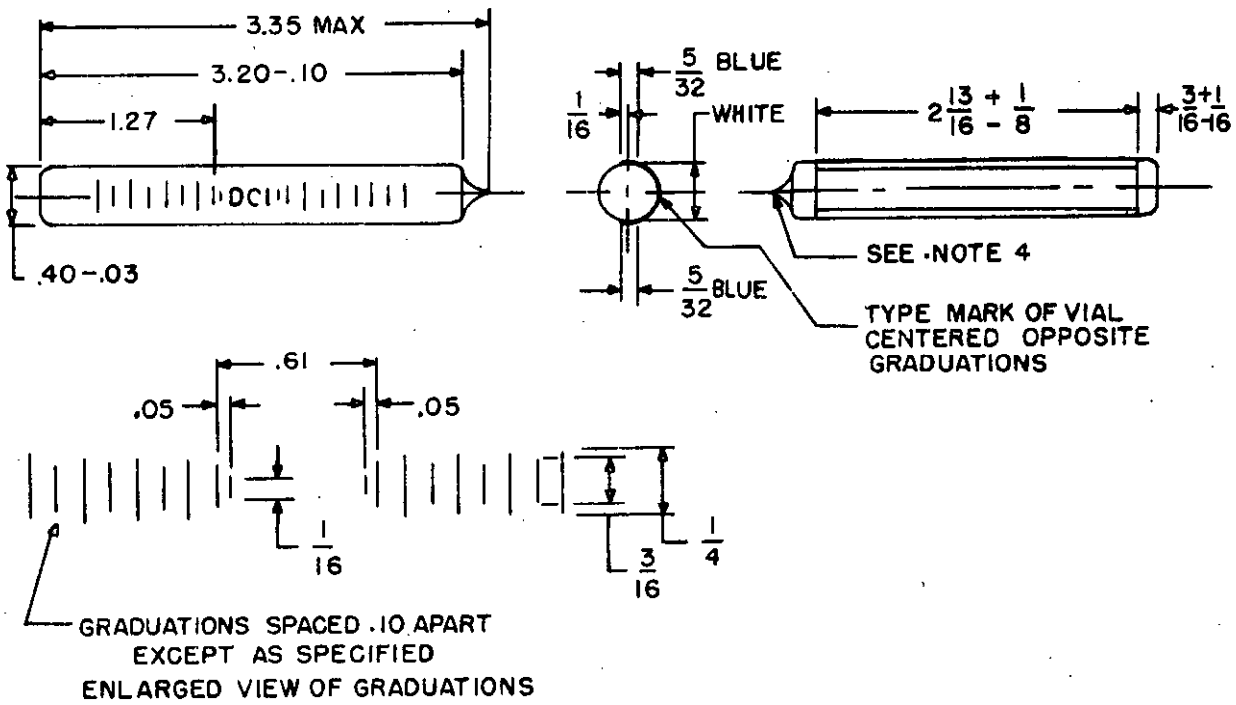
SENSITIVITY. SENSITIVITY OF LEVEL VIALS (BY TYPE) DEFINED IN MINUTES AND SECONDS (' & ") OF ARC PER ONE TENTH INCH BUBBLE TRAVEL SHALL BE AS FOLLOWS:

TYPE	SENSITIVITY	TOLERANCE MAX	TOLERANCE MIN
DB1	0' 30"	0' 20"	0' 40"
DB 2	1' 00"	0' 40"	1' 20"
DB 3	2' 30"	2' 00"	3' 00"
DB 4	5' 00"	4' 30"	5' 30"
DB 5	25' 00"	20' 00"	30' 00"

NOTES:
1. TYPE MARK TO BE 1/16 HIGH, COLOR RED
2. GRADUATIONS TO BE .015 +.005 THICK TO THE LENGTHS SPECIFIED, COLOR RED
3. BUBBLE LENGTH .60 ± .02
4. CONSTRICTION OF TUBE OPTICAL. EXTENDED TIP PERMISSIBLE TO 1.75 MAX OR FLUSH TO 1.60 DIMENSION
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCE: DECIMAL ±.010, FRACTIONAL ± 1/64

FIGURE 2. VIAL, LEVEL, CYLINDRICAL, TYPE DB

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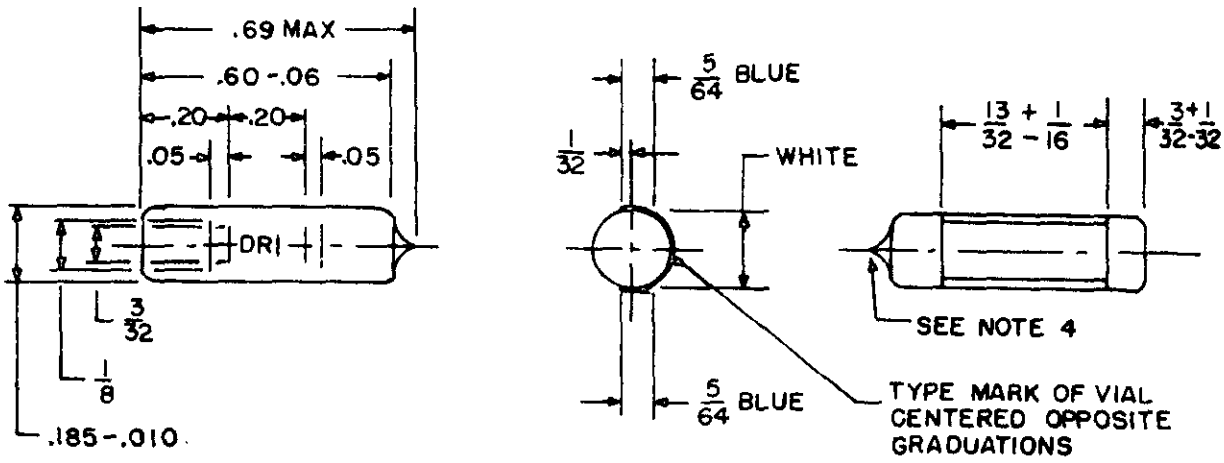
SENSITIVITY. SENSITIVITY OF LEVEL VIALS (BY TYPE) DEFINED IN MINUTES AND SECONDS (' & ") OF ARC PER ONE TENTH INCH BUBBLE TRAVEL SHALL BE AS FOLLOWS:

TYPE	SENSITIVITY	TOLERANCE MAX	TOLERANCE MIN
DC1	0' 10"	0'07"	0'13"
DC2	0' 30"	0'20"	0'40"
DC3	1' 00"	0'40"	1'20"

NOTES:
1. TYPE MARK TO BE 1/16 HIGH, COLOR RED
2. GRADUATIONS TO BE .015 ± .005 THICK TO THE LENGTHS SPECIFIED, COLOR RED
3. BUBBLE LENGTH 1.20 ± .03
4. CONSTRICTION OF TUBE OPTIONAL. EXTENTED TIP PERMISSIBLE TO 3.35 MAX OR FLUSH TO 3.20 DIMENSION
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCE: DECIMAL ± .010, FRACTIONAL ± 1/64

FIGURE 3. VIAL, LEVEL, CYLINDRICAL, TYPE DC

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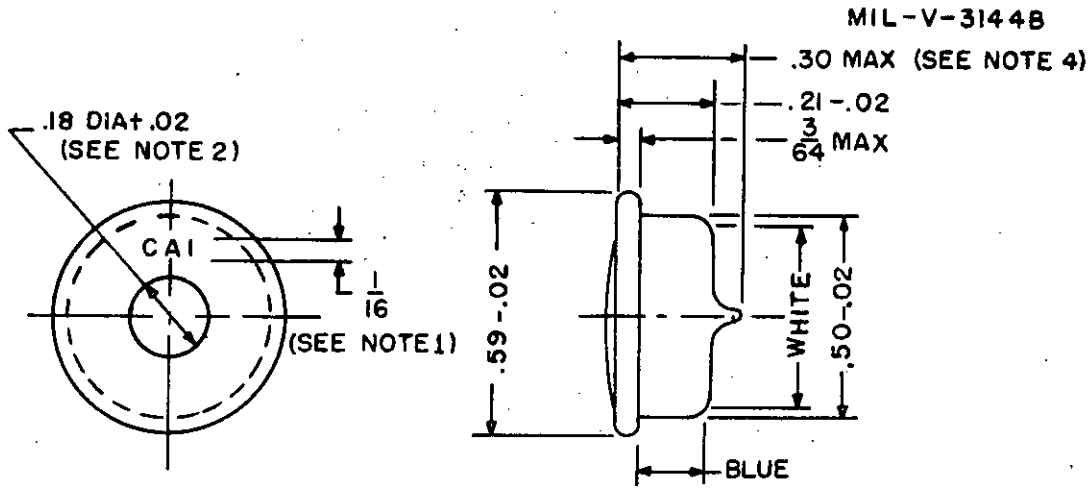
SENSITIVITY.-SENSITIVITY OF LEVEL VIALS (BY TYPE) DEFINED IN MINUTES AND SECONDS (' & ") OF ARC PER ONE TENTH INCH BUBBLE TRAVEL SHALL BE AS FOLLOWS:

TYPE	SENSITIVITY	TOLERANCE	
		MAX	MIN
DRI	70' 00"	40'00"	100'00"

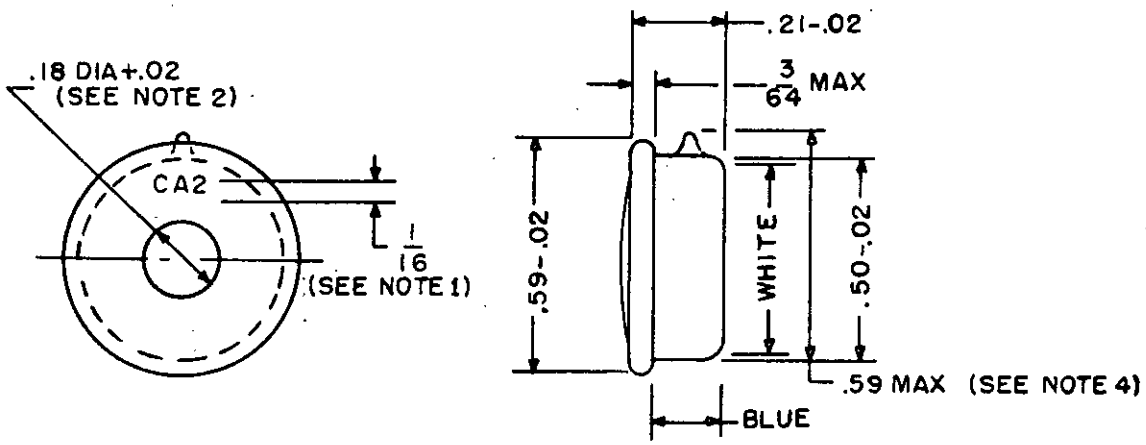
NOTES:
1. TYPE MARK TO BE 3/64 HIGH, COLOR RED
2. GRADUATIONS TO BE .015 ±.005 THICK TO THE LENGTHS SPECIFIED, COLOR RED
3. BUBBLE LENGTH .200 ± .015
4. CONSTRICTION OF TUBE OPTIONAL. EXTENDED TIP PERMISSIBLE TO .69 MAX OR FLUSH TO .60 DIMENSION
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCE: DECIMAL ± .010, FRACTIONAL 1/64

FIGURE 4. VIAL, LEVEL, CYLINDRICAL, TYPE DR

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TYPE CA1



TYPE CA2

SENSITIVITY SENSITIVITY OF LEVEL VIALS (BY TYPE) DEFINED IN MINUTES AND SECONDS (' &") OF ARC PER ONE TENTH INCH BUBBLE TRAVEL SHALL BE AS FOLLOWS:

TYPE	SENSITIVITY	TOLERANCE MAX	TOLERANCE MIN
CA1	35' 00"	10' 00"	60' 00"
CA2	17' 30"	5' 00"	30' 00"

NOTES:
1. TYPE MARK ON BOTTOM OF VIAL .015 + .005 THICK, COLOR RED
2. CIRCULAR LINE .015 + .005 THICK, COLOR RED
3. BUBBLE DIAMETER .17 ± .01
4. CONSTRICTION OPTIONAL. EXTENDED TIP PERMISSIBLE TO DIMENSION SPECIFIED OR FLUSH
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCE: DECIMAL ± .010, FRACTIONAL ± 1/64

FIGURE 5. VIAL, LEVEL, CIRCULAR, TYPE CA

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of this specification and associated drawings at a government laboratory or such other facility specified in the contract.

4.2.1.3 Initial production sample failure.- Failure of the sample to meet all requirements shall result in sample disapproval.

4.3 Quality conformance inspections.

4.3.1 Submission of product.- Unless otherwise specified by the contract, inspection lot size, lot formation and presentation for acceptance shall be in accordance with MIL-STD-105.

4.3.2 Examination and tests.- Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and Tables I, II, and III specified herein.

Table I, Classification of Defects

Class	Requirement	Test Procedure
Critical: None defined		
Major: AQL 1.0% defective		
101. Cracked, chipped or peeled	3.2, 3.4.1	4.4.1 thru 4.4.4
Minor: None defined		

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Table II, Classification of Defects

(Tests conducted at Temperature 60°F to 90°F)

Class	Requirement	Test Procedure
Critical: None defined		
Major: AQL 0.65% defective		
102. Coalescence of bubble	3.4.2	4.4.2
103. Markings water soluble	3.4.4.1	4.4.5
104. Bubble movement	3.4.4.2	4.4.6.1
105. Bubble size	3.4.4.2	4.4.6.2
106. Bubble displacement	3.4.4.3	4.4.7
107. Sensitivity (Max)	3.4.4.4	4.4.6
108. Sensitivity (Min)	3.4.4.4	4.4.6
109. Markings missing or incorrect	3.3	Visual
110. Workmanship	3.5	4.4.9
Minor: AQL 4.0% defective		
201. Overall length	3.3	Gage
202. Outside diameter	3.3	Gage
203. Spacing of graduations	3.3	4.4.8
204. Length of graduations	3.3	4.4.8
205. Width of graduations	3.3	4.4.8
206. Location of background marking	3.3	Visual
207. Color of markings	3.3	Visual

Table III, Classification of Defects

Class	Requirement	Test Procedure
Critical: None defined		
Major: AQL 1.0% defective		
111. Cracked (-65°F)	3.2	4.4.3
112. Cracked (-80°F)	3.2	4.4.4
113. Coalescence of bubble (-65°F)	3.4.2	4.4.3
114. Circular vial bubble contact (-65°F)	3.4.2.1	4.4.3.1
115. Precipitate or solid (-80°F)	3.4.3	4.4.4
Minor: None defined		

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4.4 Test methods and procedures.

4.4.1 Boiling water bath.- Vials shall be immersed in water which is brought up to boiling point and maintained for the time period specified in 3.4.1. At the end of this period and while still at this temperature extreme the vials shall be subjected to a visual examination to determine compliance with the requirements of 3.2 and 3.4.4.1.

4.4.2 Coalescence (60°F to 90°F).- Subsequent to the test of 4.4.1, and the vials having attained thermal stability at room temperature (60°F to 90°F) the vials shall be visually examined for compliance with the requirements of 3.2, 3.4.4.1 and shall then be held in a horizontal position and vigorously shaken in a vertical plane for a minimum period of 15 seconds and visually examined for compliance with the requirement of 3.4.2.

4.4.3 Coalescence (-65°F).- Subsequent to the test of 4.4.2, the same vials shall be subjected to a temperature of -65°F and maintained at this temperature for the time period specified in 3.4.1. At the end of this period and while still at this temperature extreme the vials shall be visually examined to determine compliance with 3.2, 3.4.4.1 and then held in a horizontal position and vigorously shaken in a vertical plane for a minimum period of 15 seconds and visually examined for compliance with the requirement of 3.4.2.

4.4.3.1 Circular vial at -65°F.- In conjunction with the test of 4.4.3, circular vials shall be visually examined to determine compliance with the requirement of 3.4.2.1.

4.4.4 Precipitate or solid (-80°F).- Subsequent to the test of 4.4.3, the same vials shall be subjected to a temperature of -80°F and maintained at this temperature for the time specified in 3.4.1. At the end of this period and while still at this temperature extreme the vials shall be visually examined for compliance with the requirements of 3.2, 3.4.3 and 3.4.4.1.

4.4.5 Markings water soluble.- Subsequent to the test of 4.4.4, and the vials having attained thermal stability at room temperature (60°F to 90°F), the vials shall be immersed for a period of 24 hours in a solution of water and common table salt (sodium chloride). The mixture shall contain 6 ounces of salt per gallon of water. Upon removal the vials shall be wiped with a dry cloth and visually examined for compliance with the requirement of 3.4.4.1.

4.4.6 Sensitivity.- The vial shall be placed on testing equipment capable of measuring differences of level within an accuracy of one second of arc. The angle of tilt necessary to move the bubble in the vial in increments of 0.1 inch is the basis for vial sensitivity as measured on the test device. The vial shall be visually examined to determine compliance with the requirement of 3.4.4.4.

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4.4.6.1 Bubble movement.- In conjunction with test of 4.4.6, the bubble shall be observed for compliance with the applicable requirement of 3.4.4.2.

4.4.6.2 Bubble size.- In conjunction with test of 4.4.6, the bubble size shall be measured by comparison utilizing a comparator with a magnification of at least 3X. The comparator shall contain a reticle calibrated scale to the approximate size as a means to determine compliance with the applicable requirement of 3.4.4.2.

4.4.7 Bubble displacement.- Conduct this test by means of a "V" block clamped on an angle plate or dividing head. Lay the vial lightly in the "V" block with the straight end of the vial against the closed end of the "V" block. Offset the "V" block 45° from each side of the vertical centerline to determine compliance with the requirement of 3.4.4.3.

4.4.8 Spacing, length and width of graduation lines.- Spacing, length and width of graduation lines shall be measured by comparison utilizing a comparator with a magnification of at least 3X. The comparator shall contain a reticle calibrated scale as a means to determine compliance with the requirements of 3.3 and 3.4.4.1.

4.4.9 Workmanship.- Exclusive of the characteristics specified in the classification of defects the vials shall be free from any damage, defect or contamination the nature, degree or extent of which detrimentally effects the intended use in the completed level vial.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking - Level C.- Packaging and packing shall be in accordance with commercial practice suitable for acceptance by common carrier insuring safe transportation at the lowest cost to the point of delivery.

6. NOTES

6.1 Intended use.- Glass vials covered by this specification are intended to be used as the basic element in the assembly of level vials. Level vials are used as indicating devices in the process of leveling, adjusting and setting of fire control instruments, gun mounts and similar equipment.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable type required.
(c) If bubble displacement is a requirement. (3.4.4.3)

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Custodians:
Army - MU
Navy - OS
Air Force - 84

Review activities:
Army - MU
Navy - OS
Air Force - 84

User activities:
Army - MI, WC
Navy - MC
Air Force - None

Preparing activity:
Army - MU (FA)

Categorization Code "C"

Project No. 1290-0154

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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